

Fundamentals of Optical Waveguide Fibers

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A brief explanation of optical waveguide fibers and how they affect signal transmission

OPTICAL WAVEGUIDE fibers have unique characteristics that make them attractive for communications. Their low transmission losses and high bandwidths are particularly important for long-haul, high-speed telecommunications. They also offer other advantages, including immunity to electromagnetic interference, small size, light weight, no need for electrical transmission, and excellent security against unauthorized "tapping." To make the most effective use of optical fibers, it is essential to understand the characteristics of fibers and how they affect signal transmission.

Several types of optical waveguide fibers are available or in development, with characteristics that differ in important ways. Looking back into history, the first communication-grade fibers had light-carrying core diameters on the order of the wavelength of light (a few micrometers in diameter), and could carry light in only a single waveguide mode. The concern that the difficulty of coupling light into such a small core could lead to unacceptably high losses, led to development of fibers with cores several tens of micrometers in diameter or larger, able to support many waveguide modes. Such fibers were used in the first commercial fiber-optic systems, along with semiconductor laser light sources of gallium aluminum arsenide (GaAlAs) emitting at 0.8 to 0.9 μm wavelengths.

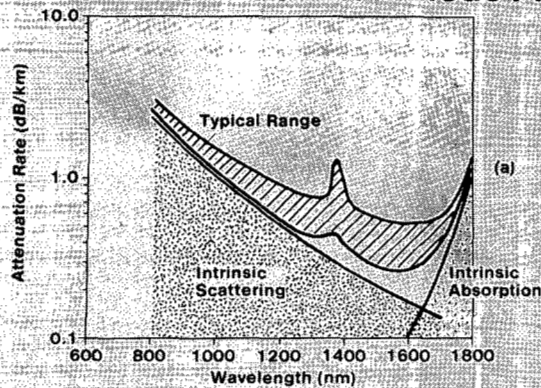
Meanwhile, further research showed that lower losses and higher bandwidth would be possible by moving to a longer wavelength, the 1.3 μm "second window," requiring indium gallium arsenide phosphide (InGaAsP) semiconductor lasers. Improvements in coupling technology and the prospect of much higher bandwidth soon made single-mode fibers practical in fourth-generation systems, which are finding widespread use as high-capacity trunk links. Meanwhile, major strides have been made toward fifth-generation fibers, single-mode types with very high bandwidth at 1.55 μm , a wavelength where fiber loss is even lower than at 1.3 μm . The new developments are not making old fiber types instantly obsolete. Instead, the different types are finding different applications, such as in short-haul links where ease of connection and low cost of components are more important than maximum transmission distance and bandwidth.

Fiber Characteristics

Three major performance criteria affect the choice of optical waveguide fibers: signal losses, ease of light coupling and interconnection, and bandwidth. All three are dependent on characteristics established in fiber design and production. Signal losses depend on attenuation within the fiber. Ease of coupling and interconnection depends on the fiber's core diameter and its numerical aperture, a measure of the angle over which light is accepted. Bandwidth depends on dispersion, a phenomenon that causes light that is originally concentrated into a short pulse to spread out into a broader pulse as it travels along an optical fiber. All three factors depend on design of the fiber, and enter into the selection of the best fiber for a particular application. Each deserves some further explanation.

Absorption and scattering of light traveling through a high-silicon-dioxide-based fiber leads to attenuation (or loss), the rate of which is measured in decibels per kilometer (dB/km). As can be seen from Fig. 1, for both multimode and single-mode fibers, attenuation depends strongly on wavelength. The decrease in scattering losses with increasing wavelength becomes offset by an increase in material absorption, so attenuation is lowest near 1.55 μm . Attenuations as low as 0.16 dB/km have been measured in laboratory-made fibers at that wavelength. However, that wavelength has yet to find commercial use because of limited bandwidth of current fibers and problems in manufacture of 1.55- μm lasers. Current high-capacity, long-distance links use single-mode fibers at 1.3 μm , despite somewhat higher attenuation, because bandwidth is much larger and lasers are available commercially. If low component cost is more important than high

Graded-Index Multimode Fibers



Single-Mode Fibers

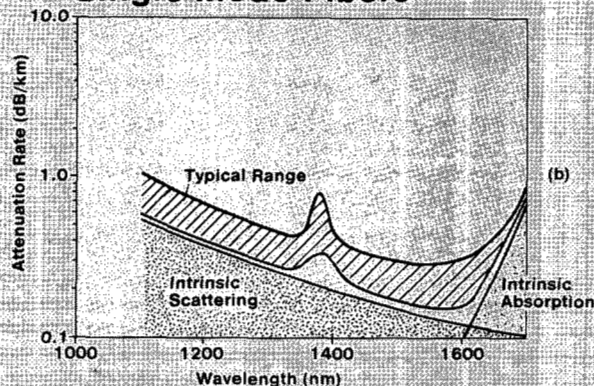


Fig. 1a & 1b. Spectral attenuation rate of (a) multimode graded-index and (b) single-mode fibers. Fiber attenuation rate is intrinsically limited by increased scattering toward short wavelengths and absorption at lower wavelengths. The absorption band at 1390 nm is due to a small hydroxyl impurity in the glass. The typical attenuation rate range for commercially available fibers is shown.

capacity and long-haul operation, fiber-optic systems can operate in the 0.8 to 0.9 μm "first window," using GaAlAs lasers that are much less costly than the InGaAsP lasers needed at 1.3 μm . There, multimode fiber loss rates are typically 2.7 to 2.8 dB/km.

Optical fibers normally are manufactured in lengths of two kilometers or more, which are then housed in cable structures and spliced together. Proper cabling causes minimal extra loss, but poor cable structures can cause "microbending" of the fiber, which increases attenuation by making light leak out. Each good splice adds a small fraction of a decibel of extra loss. Demountable connectors have somewhat higher losses than permanent splices with precise values dependent on fiber and connector type.

The size of the fiber core is generally related to the ease of coupling light from any source into a fiber and for interconnecting fibers. Mechanical tolerances are more stringent for small-core fibers, particularly for single-mode fibers, which have core diameters around 10 μm . Generally speaking, if the fiber is to collect light efficiently, its core area should be no smaller than the emitting area of the light source. This means that for single-mode fibers, the light sources should be lasers emitting in a single transverse mode. Multimode fibers, with core diameters ranging from 50 to as large as 1000 μm , can collect output from light-

emitting diodes (LED's), which have much larger emitting areas, are much less expensive, and have high reliability.

A second factor which affects ease of coupling and interconnection is the fiber's numerical aperture. This quantity is a measure of the fiber's angular light-gathering ability. Using a simple ray model and in keeping with a basic optics definition, the numerical aperture (NA) of the fiber is defined as the sign of the vertex angle, θ_c , of the largest cone of meridional rays that can enter or leave the fiber.

This measures the acceptance angle of the fiber, as shown in Fig. 2. Light rays that fall within this acceptance angle are guided within the fiber; those that enter outside this angle are not confined to the fiber, and thus leak out or radiate away and are lost. Fiber NA can be calculated if the variation of refractive index as a function of distance from the fiber axis is known, but normally it is easier to directly measure the NA because of other practical considerations.

The information-carrying capacity or bandwidth of a fiber is inversely related to its total dispersion. The total dispersion is a combination of three components: mode dispersion, material dispersion, and waveguide dispersion. Mode dispersion is dependent only on the fiber, while the amount of material and waveguide dispersion depends on the breadth of the wavelength spectrum being emitted by the light source.

Mode dispersion occurs in multimode fibers because different modes travel different effective distances through the optical fiber. A simple graphic example, for step-index multimode fiber is shown in Fig. 3a, where the light ray traveling along the center of the fiber has a shorter path than the one zig-zagging back and forth. This causes light in the different modes to spread out temporally as it travels along the fiber, and the spreading continues as the light pulse travels farther. For multimode step-index fibers, this effect can limit the step-index bandwidth to the order of 20 megahertz for a one-kilometer length, or 20 MHz-km.

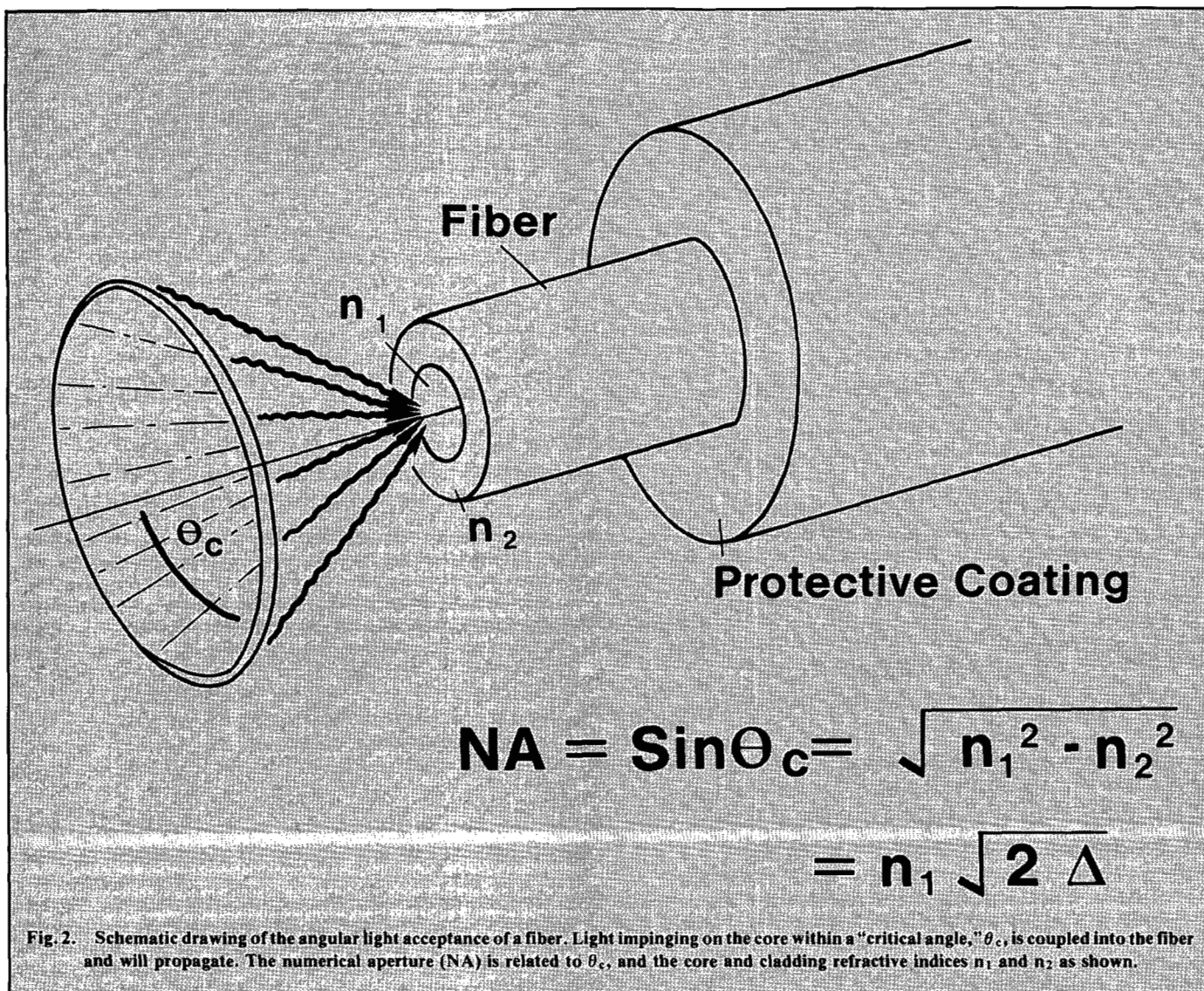
Multimode dispersion can be reduced by making the refractive index of the fiber core graded in the radial direction rather than constant, hence the name graded-index fibers. As shown in Fig. 3b, the rays in a graded-index fiber follow sinusoidal paths. Even though a high-angle ray travels a longer distance than a low-angle ray, much of its path is in a low refractive-index region where it travels more rapidly. By careful design of the index profile, nearly equal transit times for all rays is possible. Bandwidths of around a GHz-km are readily attainable in commercial graded-index fibers.

A third alternative for removing multimode dispersion is to use a single-mode fiber. As the name implies, since there is only a single mode, no dispersion between the modes can exist and very high bandwidths are possible. The bandwidth of the single-mode fiber, however, is not infinite because of the other two sources of dispersion mentioned earlier—material and waveguide dispersion.

Material and waveguide dispersion are subtly different effects that arise because the way light propagates through materials is a function of wavelength. Both are measured in picoseconds (of pulse spreading) per nanometer (of source spectral width) per kilometer (of fiber length), reflecting both the increase in magnitude with source linewidth, and the accumulation of dispersion over distance in a fiber. The sum of these two quantities is the total dispersion for a single-mode fiber.

Material dispersion is caused by the variation in refractive index of glass with wavelength. (It is sometimes called spectral or chromatic dispersion because this is the same effect which causes a prism to spread out a spectrum, but those terms are often misused and best avoided.) This leads to pulse spreading even when different wavelengths follow the same path, because the speed of light in the fiber equals the speed of light in a vacuum scaled by the index of refraction at a given wavelength.

Waveguide dispersion occurs because light in a single-mode fiber is not confined completely to the core; about 20 percent of the light travels in the cladding adjacent to a step-index core. The



refractive index of the cladding is lower than that of the core and so the light wants to travel somewhat faster than in the core.

Since the core and cladding light belongs to the same mode, however, it must travel at the same speed and does so at an effective velocity between that of the core and cladding materials. The exact value of the effective velocity changes with wavelength, so waveguide dispersion is also wavelength-dependent, although the change in waveguide dispersion with wavelength is smaller than that of material dispersion, as shown in Fig. 4. The amount of waveguide dispersion depends on the design of the waveguide structure as well as on the fiber materials.

The material and waveguide dispersion can have different signs, and thus completely cancel each other out. In conventional germanium-doped silica fibers, this "zero-dispersion" wavelength is near $1.3 \mu\text{m}$. Strictly speaking, the bandwidth of single-mode fibers does not become infinite at that wavelength, but it does go beyond most practical requirements. The zero-dispersion wavelength is also important for graded-index fibers because it removes the increase in dispersion with an increase in spectral bandwidth of the source. Thus, it is possible to use spectrally broad-band LED's at $1.3 \mu\text{m}$ without the high dispersion that their linewidth would cause at other wavelengths.

The interaction of dispersion, ease of coupling and interconnection, and attenuation rate determine the practical use-

fulness as a fiber. By picking different combinations of these characteristics, different functional needs can be fulfilled. Fiber makers are also working on ways to combine desirable characteristics, notably the low dispersion at $1.3 \mu\text{m}$ and the low attenuation at $1.55 \mu\text{m}$, in new fiber types.

Types of Fibers and Applications

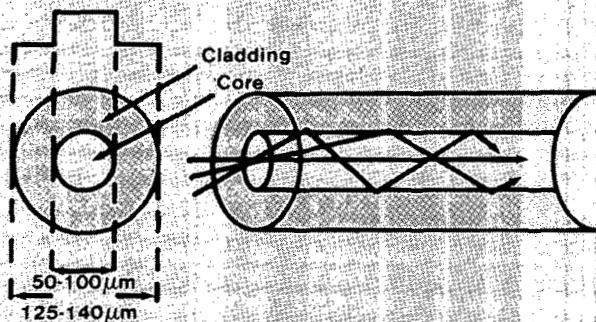
A wide range of optical waveguide designs are possible in theory, and many have been made in the laboratory. No one design meets all application needs. Manufacturers have concentrated on three broad classes: single-mode, graded-index multimode, and step-index multimode.

The single-mode fiber, by definition, is one that carries light in only a single waveguide mode. Traditionally, the internal structure of single-mode fibers has been simple—a core of high-index glass surrounded by a much thicker cladding of lower-index glass, separated by an abrupt boundary. However, new designs in the works use more complex structures to improve performance, as described later in this article.

An optical fiber with core diameter $2a$, a homogeneous core refractive index, a fractional index difference between core and cladding $\Delta = \frac{\Delta n}{n}$, will transmit a single mode for all wave-

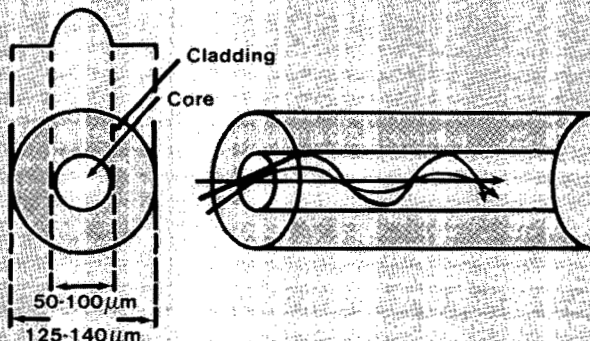
Multimode Fiber

Refractive Index Profile



Step Index (a)

Refractive Index Profile



Graded Index (b)

Fig. 3. Schematic illustration of the propagation of light rays in multimode (a) step- and (b) graded-index fibers. In step-index fibers the rays may be thought to follow a zig-zag path while in graded-index fibers the paths are sinusoidal.

lengths longer than the cutoff wavelength λ_c .

$$\lambda_c = \frac{2\pi a n \sqrt{2\Delta}}{2.405}$$

At shorter wavelengths, the fiber will support two or more modes. As this indicates, the core of a single-mode fiber must be small, roughly a few times the cutoff wavelength, or several μm 's for operation at 1.3 or 1.55 μm . The design leads to tight mechanical tolerances in coupling, and to a numerical aperture smaller than other types, but transmission of a single mode gives very broad bandwidth for signals.

Single-Mode Step-Index Dispersion

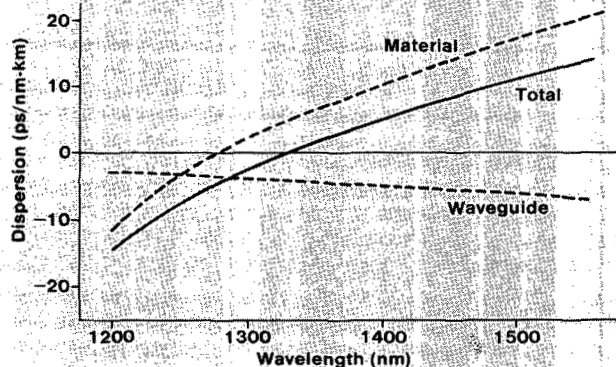


Fig. 4. The total dispersion of single-mode fibers is a combination of material and waveguide dispersion. This produces a zero-dispersion point near 1330 nm where extremely large bandwidth is possible.

Graded-index multimode fibers were developed in the early 1970s as a compromise between the high bandwidth of single-mode fibers and the easy coupling of step-index multimode fibers. The core in such fibers is radially graded in composition, rather than being abrupt, in such a way that the path length differences

Single-Mode Designs

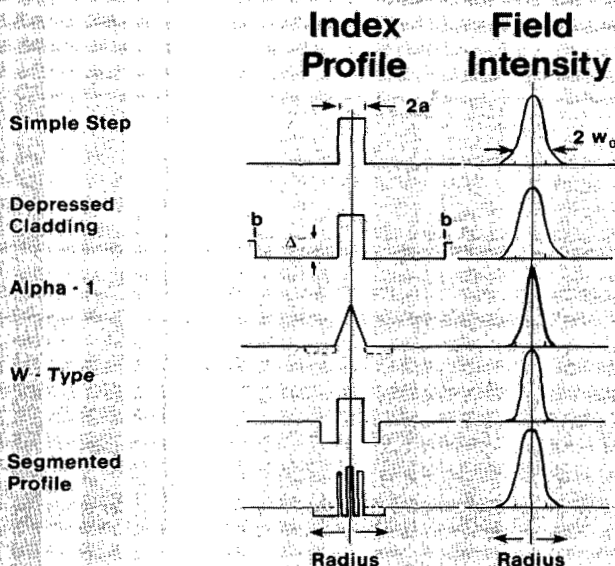


Fig. 5. Radial refractive index profiles and light field intensities for several special single-mode fibers under recent investigation. These designs alter the dispersion characteristics of the single-mode fiber.

between refracting light rays is minimized. This reduces multimode dispersion greatly, allowing transmission bandwidths greater than one gigahertz-kilometer for fibers with 50- μm cores.

Graded-index multimode fibers with 50- μm cores and 125- μm cladding have been used in many telecommunications systems operating at 45–140 megabits per second for distances in the 10–20-km range. For transmission at higher data rates or over longer distances, single-mode fibers are now used. A new graded-index fiber with 85- μm core has been developed for short-distance applications requiring easy coupling and high data rates, such as video and high-speed digital local-area networks. The 85- μm core permits significantly better input coupling than the smaller 50- μm core developed for telecommunications, without a significant degradation in bandwidth.

Step-index or quasi-step-index multimode fibers have core diameters from about 50–1000 μm , with a 100- μm core/140- μm cladding design having become standard for some short-distance applications. The large core diameter and high numerical aperture of step-index multimode fibers simplifies input coupling, and allows use of inexpensive connectors. It also permits use of inexpensive LED light sources. However, strong multimode dispersion limits transmission bandwidth.

Different types of step-index multimode fiber are available: glass-clad glass fibers, plastic-clad silica, or all-plastic versions, in various diameters. In glass-on-glass fibers, core and cladding are made of glasses with different compositions, chosen to provide optical waveguiding in the fiber. In plastic-clad silica fibers, the core is pure silica, clad with a plastic that has lower refractive index.

An alternative is the all-plastic fiber, with core and cladding having different refractive indexes. Each structure has advantages and disadvantages; all-plastic fibers, for example, are easy to handle but may have losses typically in the 100 dB/km range. Increases in light-coupling efficiency that come with larger core diameter, come with a price of greater material requirements and reduced bending flexibility of the fiber. The relatively high loss of all-plastic fibers limits them to very short-distance systems, such as transmitting signals throughout the interior of an automobile.

Emerging Types of Single-mode Fiber

Present single-mode, step-index fibers permit transmission above 400 Mb/s with repeater spacings of tens of kilometers. Such fiber-optic links are vital components of the new national long-haul telecommunication network emerging from the deregulation of long-distance telephony. However, widespread interest remains in transmission at even higher data rates over long distances, to increase the growing volume of telecommunications traffic both on land and between continents. This interest, plus special-purpose applications such as sensing, have led to new types of single-mode fibers.

Maximum data rate and transmission distance of present single-mode fibers and laser sources are sub-optimized because the zero-dispersion wavelength falls at 1.3 μm , while the minimum loss is at 1.55 μm . At low data rates, maximum transmission distance at 1.55 μm is limited by loss, but at high data rates the limit comes from dispersion, which causes an overlap between successive pulse slots. Because dispersion is proportional to source spectral width, the optimization can be approached either by trying to narrow the laser spectral width, or by lowering fiber dispersion at 1.55 μm .

Notable progress has been made in narrowing laser spectral width, by investigating laser structures which tend to stabilize emission wavelength. The leading approach now is the distributed-feedback laser, in which a grating in the semiconductor material reflects light of a certain wavelength back to the laser's active region, thus stabilizing output in a narrow range of wavelengths. The current record for data rate and distance was obtained using such a laser, in which researchers at AT&T Bell Laboratories

transmitted 4 Gb/s through 103-km of single-mode fiber without any repeaters. This record tends to be broken every few months. However, direct modulation of the laser with the signal sets an inherent lower limit on the degree of stabilization possible. Injection of current carriers into the semiconductor laser changes its refractive index, and thus the wavelength of light it generates as it emits a pulse of light—a phenomenon called “chirp.” In addition, there remain unanswered questions about the manufacturability of 1.55- μm semiconductor lasers, particularly with the complex structures required to limit emission linewidth.

The alternative is to reduce fiber dispersion, either by shifting the zero-dispersion wavelength to 1.55 μm , or by spreading the dispersion minimum out over a broader range. (The intrinsic scattering limit in glass fibers, shown in Fig. 1b, makes it impossible to achieve as low loss at 1.3 μm as at 1.55 μm .) The basic material dispersion is hard to alter significantly, but it is possible to alter the waveguide dispersion by moving from a simple step-index design to more complicated profiles such as those shown in Fig. 5. Addition of material and waveguide dispersion then can produce zero dispersion at 1.55 μm , or can result in very low dispersion over a range of wavelengths between 1.3 and 1.55 μm , approaches known as dispersion-shifted and dispersion-flattened fibers respectively.

Dispersion-shifted fibers are the simpler to produce, because parameters of the fiber design have to be adjusted to optimize dispersion only at one wavelength. Some early dispersion-shifted fibers had higher attenuation at 1.55 μm than conventional types, but our recent results indicate that dispersion-shifted fiber can be made without any significant increase in attenuation at 1.55 μm . Such fibers have now reached the market, and should permit high data rate transmission over still longer distances than the present step-index single-mode fibers.

Dispersion-flattened fibers are more complex to design, because dispersion must be considered over a range of wavelengths. As for dispersion-shifted fibers, there has been some concern that higher losses would accompany the change in dispersion. We have produced dispersion-flattened fibers at Corning Glass Works, and although detailed work is still in progress, it seems that they, too, can be made with losses only marginally higher than those of conventional fibers, even at 1.55 μm .

Maximum data rate and transmission distance with dispersion-flattened fibers would not quite match that of dispersion-shifted fibers at 1.55 μm . However, dispersion-flattened fibers offer desirable characteristics over a much broader range of wavelengths, and thus could be used for wavelength-division multiplexing (WDM). One possibility would be installation of such fiber for operation at one wavelength, keeping open the possibility of adding additional transmission wavelengths to boost the fiber capacity. Another possibility is design of systems explicitly for wavelength-division multiplexing over dispersion-flattened fibers. So far, wavelength-division multiplexing has not gained wide acceptance in commercial systems.

Polarization-maintaining Fiber

Strictly speaking, single-mode fibers are not single-moded. They, in fact, support two lowest-order modes with orthogonal linear polarizations. In fibers that are circularly symmetric about their axis, those two modes are degenerate (that is, light in them travels at the same speed), and light can shift between them. This does not produce insurmountable problems for present communication systems, but it does limit some concepts for fiber-optic sensing.

Fibers that discriminate between these polarization modes have been made by creating fibers with non-circularly symmetric refractive index in or near the core. Typically, this is done by fabricating a fiber with diametrically opposed stress regions extending longitudinally along the core, and thus creating the

required refractive index distribution. Some polarization-maintaining fibers have already reached the market, and could prove important in sensing applications.

Prices are presently high compared to those of conventional single-mode fibers, and the distances over which isolation of the two polarizations can be maintained are limited to a few kilometers.

Conclusions

The rapid progress in optical waveguide fiber technology has led to a proliferation of types as well as marked improvements in performance. The worst fiber today is perhaps an order of magnitude better than the best fiber of a decade ago. This healthy diversity is essential for fiber optics to find a broad range of communications applications. Just as high-bandwidth coaxial cable is not the right choice to carry a single voice channel from a telephone pole to a house, a variety of fiber types will be needed to accommodate this broad range of applications. The final choice of fiber type is likely to be dictated by the types of active and passive components available to build the system, as well as secondary issues such as installation and maintenance costs. The present diversity of fiber types makes fiber-optic transmission a viable candidate for applications ranging from a trans-Atlantic cable to interference-free transmission of signals within automobiles.

More may be coming, too. Fibers made of non-oxide glasses might eventually have even lower attenuation than today's best silicon dioxide-based glass fibers, although at longer infrared wavelengths. Their use of unusual materials could present obstacles to their widespread use in communications, but in the long term, they could find their own niche in applications ranging

from extremely long repeaterless links to transmission of infrared laser beams for industrial applications.

D. B. Keck received his B.S., M.S., and Ph.D. degrees in Physics from Michigan State University, the latter in 1967. He subsequently joined the research and development laboratory of Corning Glass Works where he is presently manager of the Applied Physics Research Department, a position he has held since 1979. Dr. Keck's research areas include molecular spectroscopy, gradient index optics, and optical fiber waveguides. In the latter area, he did some of the pioneering work on waveguide attenuation and dispersion and has been extensively involved with measurement of the optical properties of fibers. He has lectured on the subject of optical fiber communications in courses at the University of California at Santa Barbara, the University of Rochester Institute of Optics, and at George Washington University. He is a coauthor of portions of two books on the subject. Dr. Keck has served on the committees for the OSA Topical Meeting on Gradient Index Optical Imaging Systems, the International Integrated Optics and Optical Communications Conference, and the Symposium on Optical Fiber Measurements. He has served as program and general co-chairman for the Topical Meeting on Optical Fiber Communications in 1982 and 1983, as well as program chairman for GRIN V in 1984. He is presently on the International Management Committee for the 1985 International Integrated Optics and Optical Communications Conference. In 1981 he was co-recipient of the Technology Achievement Award of SPIE, the International Society for Optical Engineering, and the Industrial Research IR-100 Award and in 1983 the Engineering Materials Achievement Award of the American Society for Metals for his work on optical fibers. His professional society affiliations include: Fellow of Optical Society of America, and member of Sigma Xi and Sigma Pi Sigma. He has served as a member of the United States National Committee to the International Commission for Optics. Dr. Keck has authored more than 70 papers and publications, and holds patents. ■